



TECHNICAL UNIVERSITY OF CRETE
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

Diploma Thesis

**Classification of Dominant
Functional Connectivity Graphs
from Mild Traumatic Brain Injuries
using Kolmogorov-Arnold Networks (KAN)**

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Chania, July 2026



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ΣΧΟΛΗ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ

Διπλωματική Εργασία

Ταξινόμηση Κυρίαρχων Γραφημάτων
Λειτουργικής Συνδεσιμότητας
από Ήπιες Τραυματικές Εγκεφαλικές Κακώσεις
με Χρήση των Δικτύων Kolmogorov-Arnold (KAN)

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Χανιά, Ιούλιος 2026

Abstract

The clinical diagnosis of Mild Traumatic Brain Injury (mTBI) remains a formidable challenge due to the subtle neurophysiological changes that characterize the condition, which conventional structural neuroimaging typically fails to detect. This thesis proposes a pipeline for classifying mTBI from healthy subjects by analyzing Magnetoencephalography (MEG) data. Specifically, it combines dynamic functional connectivity microstate analysis with Kolmogorov-Arnold Networks (KAN), emphasizing interpretability and visual transparency.

The analysis was based on thresholded, time-varying, dominant functional connectivity graphs of dimensionality 90x90 (iPLV, OMST). These graphs were decomposed via normalized Laplacian eigenspectra and quantized into two prototypical network microstates using the Neural Gas algorithm. An eigenvalue-based sorting criterion was introduced to label the microstates as higher-connectivity and lower-connectivity across all subjects. Ten temporal and graph-theoretic features were extracted per subject. These features were fed into a shallow, single-layer KAN, which is an emerging architecture that replaces fixed edge weights with learnable B-spline activation functions and derives feature importance intrinsically during training. To address class imbalance, a random undersampling routine was implemented across 100 independent runs, yielding a balanced dataset of 60 subjects. Model evaluation followed a two-phase protocol: 5-fold stratified cross-validation on 48 training subjects for feature threshold selection, and Leave-One-Out testing on a held-out set of 12 subjects.

The model achieved a test accuracy of 0.707 ± 0.133 and AUC-ROC of 0.760 ± 0.141 . While both SVM and MLP slightly outperformed the network by 4–5% in accuracy under the same protocol, KAN intrinsically isolated transition probability and mean dwell time in the lower-connectivity state as the strongest mTBI biomarkers. The learned spline functions revealed non-linear, non-monotonic relationships between these features and the classification output, providing a neurophysiologically grounded explanation of the model’s decisions. This work is proof that interpretable neural architectures can provide transparent diagnostic insights with minimal impact on classification performance.

Περίληψη

Η κλινική διάγνωση της ήπιας κρανιοεγκεφαλικής κάκωσης (mTBI) αποτελεί διαρκή πρόκληση λόγω των ήπιων νευροφυσιολογικών συμπτωμάτων που τη χαρακτηρίζουν, τα οποία δεν είναι εύκολο να εντοπιστούν με μεθόδους συμβατικής δομικής νευροαπεικόνισης. Η παρούσα διπλωματική προτείνει μια μέθοδο ταξινόμησης μέσω ανάλυσης δεδομένων Μαγνητοεγκεφαλογραφήματος (MEG). Συνδυάζει την ανάλυση μικροκαταστάσεων δυναμικής λειτουργικής συνδεσιμότητας με τα Δίκτυα Kolmogorov-Arnold (KAN), δίνοντας έμφαση στην ερμηνεία και τη διαφάνεια των αποτελεσμάτων.

Η μελέτη βασίστηκε σε κατωφλιωμένους, χρονικά μεταβαλλόμενους γράφους κυρίαρχης λειτουργικής συνδεσιμότητας διάστασης 90×90 (iPLV, OMST). Από αυτούς τους γράφους εξήχθη το φάσμα ιδιοτιμών του κανονικοποιημένου πίνακα Laplace, και ακολούθησε ο κβαντισμός τους σε δύο πρότυπες καταστάσεις μέσω του αλγορίθμου (Neural Gas). Βάσει ενός κριτηρίου ταξινόμησης που στηρίχθηκε στις ιδιοτιμές Laplace, οι καταστάσεις διαχωρίστηκαν σε υψηλής και χαμηλής συνδεσιμότητας αντίστοιχα. Από τις προκύπτουσες συμβολικές χρονοσειρές εξήχθησαν δέκα χρονικά και γραφοθεωρητικά χαρακτηριστικά ανά υποκείμενο. Αυτά τα χαρακτηριστικά χρησιμοποιήθηκαν ως είσοδος σε ένα ρηχό, μονοεπίπεδο δίκτυο KAN, το οποίο είναι μια καινοτόμα αρχιτεκτονική που αντικαθιστά τα σταθερά βάρη των ακμών με learnable συναρτήσεις ενεργοποίησης B-spline και εξάγει τα πιο σημαντικά χαρακτηριστικά άμεσα κατά την εκπαίδευση. Για την αντιμετώπιση της ανισορροπίας των κλάσεων εφαρμόστηκε τυχαία υποδειγματοληψία σε 100 ανεξάρτητες επαναλήψεις, παράγοντας ένα ισορροπημένο σύνολο 60 υποκειμένων. Το μοντέλο αξιολογήθηκε μέσω ενός πρωτοκόλλου δύο φάσεων: stratified 5-fold cross-validation για την επιλογή των κυρίαρχων χαρακτηριστικών, και Leave-One-Out testing σε ένα ανεξάρτητο test dataset.

Το μοντέλο πέτυχε ακρίβεια 0.707 ± 0.133 και AUC-ROC 0.760 ± 0.141 . Αν και τα SVM και MLP ξεπέρασαν το KAN σε ακρίβεια κατά 4–5% μέσω του ίδιου πρωτοκόλλου αξιολόγησης, το KAN κατάφερε από μόνο του να απομονώσει την πιθανότητα μετάβασης και τον μέσο χρόνο παραμονής στην κατάσταση χαμηλότερης συνδεσιμότητας ως τους ισχυρότερους δείκτες mTBI. Οι συναρτήσεις spline που έμαθε το μοντέλο αποκάλυψαν μη γραμμικές, μη μονότονες σχέσεις μεταξύ αυτών των χαρακτηριστικών και της εξόδου ταξινόμησης, παρέχοντας μία νευροφυσιολογικά θεμελιωμένη ερμηνεία των αποφάσεων του μοντέλου. Η εργασία αυτή αποδεικνύει ότι τα ερμηνεύσιμα νευρωνικά δίκτυα μπορούν να προσφέρουν διαγνωστικά στοιχεία με διαφάνεια έναντι ελάχιστου κόστους στην ακρίβεια ταξινόμησης.

Acknowledgments

This thesis would not have been possible without the support of several people to whom I am truly grateful.

First, I would like to thank my supervisor, Professor Michalis Zervakis, and Assistant Professor Marios Antonakakis, for their guidance and support throughout this work. I am also thankful to the members of my examination committee, Professor Dionysios Christopoulos and Professor Athanasios Liavas, for their time and feedback.

None of this would have meant anything without my family — Georgia, Panagiotis, and Antonia. They supported me in every way that they could throughout this degree, so whatever I have accomplished here, I owe to them.

Finally, to my friends — those I met during School and those who were already there long before — thank you for the memories we built together over these years and for always showing up when it mattered. I hope that graduation is not an ending but simply the next chapter, and that there are many more memories ahead of us.

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Chapter 1 – Introduction

1.1 Motivation

Mild Traumatic Brain Injury (mTBI) is often referred to as a “silent epidemic” since it affects a very large number of people who suffer from its cognitive and functional consequences. Yet, it remains one of the most difficult neurological conditions to diagnose with absolute certainty. Conventional structural neuroimaging, such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) scans, is frequently unable to detect the subtle, functional abnormalities that characterize a “concussed” brain. As a result, clinical assessment often relies on subjective self-reporting and neuropsychological tests, which can be time-consuming and, in most cases, inconsistent. Consequently, the need for objective, data-driven biomarkers that can identify the functional “fingerprints” of mTBI is now more essential than ever.

Magnetoencephalography (MEG) is a cutting-edge technology that can address this need. Its high temporal resolution can capture rapid changes in brain activity, while resting-state recordings allow us to observe the brain’s intrinsic organization without the distraction of a specific task. Within this framework, dynamic functional connectivity (dFC) analysis goes a step further than static approaches by treating brain activity as a time-varying signal. This reflects how the brain spontaneously transitions between different functional states, known as “network microstates.” Previous research has shown that features derived from these microstates can distinguish mTBI patients from healthy controls with high accuracy. These findings suggest that mTBI affects not only functional topology but also the stability of these microstate transitions, which is the primary motivation for the feature design in this thesis.

However, an important methodological gap remains in many dFC pipelines: the semantic consistency of state labels across subjects. When a label is assigned to a connectivity prototype by a quantization algorithm, that label is mathematically arbitrary. This means that “State 1”, for example, might look completely different from one subject to another. Without a principled method of anchoring these labels, comparing groups of state-specific features can be unreliable. Addressing this gap is an important part of this work.

After consistent dynamic features are extracted, the next challenge is classification. One

constraint is that clinical MEG datasets are typically small, which makes standard deep learning difficult. Another concern is the need for the models used by clinicians as decision-support tools to be explainable and transparent. Both these demands are addressed by Kolmogorov–Arnold Networks (KANs), a recently proposed alternative to Multi-Layer Perceptrons (MLPs). By replacing fixed node activations with learnable spline functions on the edges, KANs are more parameter-efficient compared to MLPs, especially with small datasets. They also allow the decision-making process to be visualized directly. Feature importance is interpreted by the model’s own internal functions, so there is no need for external “black-box” explanation tools that are typically applied after a model is already trained.

1.2 Contributions

First Application of KANs to Neuroimaging Classification

This work is the first to apply KANs to a neuroimaging classification problem. One of the benefits of using a KAN architecture instead of an MLP-based classifier is that there is no need for independent statistical ranking to isolate biomarkers. KANs are able to estimate feature importance intrinsically during the training process. The learned spline functions provide an explicit mapping of how each dynamic feature affects the classification outcome. So, the visualization of these functions leads to a clinically transparent view of what distinguishes mTBI brain dynamics from healthy ones. This establishes the fact that high-performance predictive models do not need to function like “black boxes” that compromise interpretability in favor of predictive accuracy.

Deterministic State-Label Anchoring for Group-Level dFC Analysis

A principled, deterministic procedure for anchoring microstate labels across subjects is introduced. Because the Neural Gas algorithm runs independently per subject, the label assigned to each prototype is mathematically arbitrary. This makes group-level comparisons of state-specific chronnectomic features unreliable. The proposed anchoring method sorts each subject’s prototypes by the mean eigenvalue of their normalized Laplacian spectrum. This sorting is grounded in the established relationship between Laplacian eigenvalues and network integration (Chung and Graham, 1997), and it ensures that the higher-connectivity and lower-connectivity microstates are labeled consistently across all subjects.

1.3 Thesis Structure

Chapter 2 establishes the neurophysiological and mathematical basis of this work. It reviews the functional organization of the human brain, the pathophysiology of mild traumatic brain injury (mTBI), and Magnetoencephalography (MEG). Also, it details the transition from static to dynamic functional connectivity and states the mathematical framework of Kolmogorov-Arnold Networks (KANs).

Chapter 3 explains the previous work that was the foundation of this thesis. It covers sensor-level noise cleaning, source reconstruction via LCMV beamforming, atlas parcellation, and the extraction and Orthogonal Minimal Spanning Tree (OMST) filtering of cross-frequency dynamic functional connectivity graphs (DFCGs) to produce the ready-to-use dominant connectivity tensors.

Chapter 4 describes the primary methodological steps of this thesis. It details the Laplacian eigendecomposition of connectivity snapshots, the Vector Quantization process using the Neural Gas algorithm, and a deterministic state-label anchoring method. It also demonstrates the extraction of a ten-dimensional chronnectomic and graph-theoretic feature vector following a two-phase nested cross-validation and Leave-One-Out (LOO) evaluation protocol.

Chapter 5 presents the empirical findings of the study. It benchmarks the classification performance of the proposed KAN architecture against SVM and MLP models, evaluates the intrinsic feature importances across pruning thresholds, and provides a visual analysis of the learned B-spline activation functions.

Finally, Chapter 6 discusses the findings in the context of the existing literature, interprets the neurophysiological significance of the selected biomarkers, and acknowledges the main limitations of the present work. Suggestions for future research are also outlined.

Chapter 2 – Theoretical Background

2.1 Brain Structure and Functional Organization

The human brain is the most important part of the nervous system, responsible for coordinating all cognitive, sensory, and motor functions (Visible Body, 2023). The most external layer of the brain, the cerebral cortex, is divided into four lobes with distinct functional specializations. The frontal lobe controls all executive functions, working memory, attention, and decision-making. The parietal lobe integrates sensory information and spatial awareness. The temporal lobe is responsible for memory consolidation, language comprehension, and auditory processing. And finally, the occipital lobe is mostly dedicated to visual processing (Mayfield Clinic, 2023; Johns Hopkins Medicine, 2023).

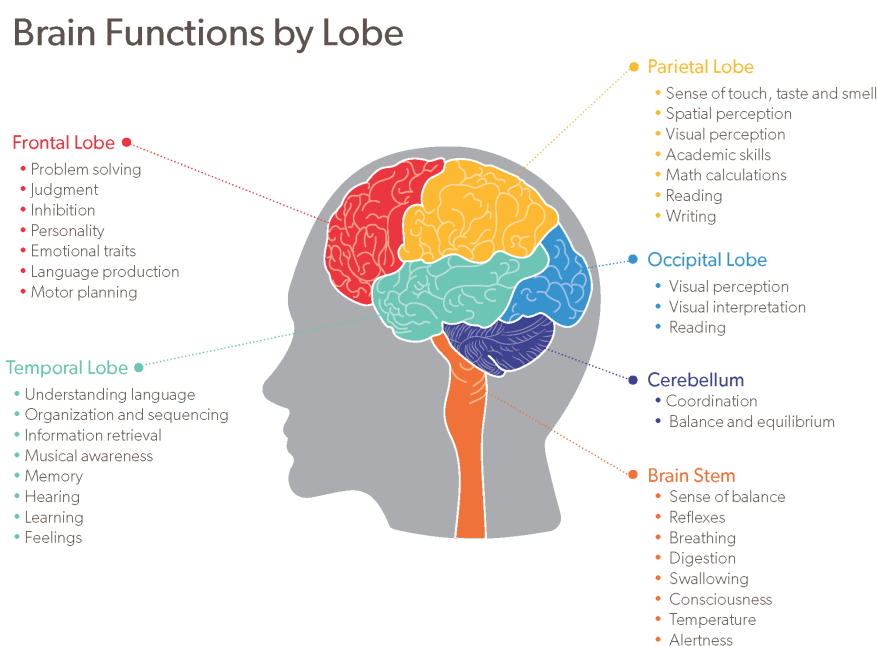


Figure 2.1: The four lobes of the cerebral cortex and their primary functionalities. Adapted from Shirley Ryan AbilityLab (2023).

What makes the brain remarkable from a systems perspective is that these areas do not function independently. Distributed neural assemblies are interconnected through white matter pathways and communicate via the synchronization of oscillatory activity

(von der Malsburg, 1981; Gray and Singer, 1989). To be more precise, populations of neurons across different regions fire in coordinated rhythms, establishing functional links that support cognition. This synchronization occurs across several frequency bands; delta (δ , 0.5–4 Hz), theta (θ , 4–8 Hz), alpha (α , 8–13 Hz), beta (β , 13–30 Hz), low gamma (γ_{low} , 30–55 Hz), and high gamma (γ_{high} , 55–90 Hz). Each band is associated with distinct cognitive processes — theta with memory encoding, alpha with attentional gating, beta with sensorimotor processing and top-down control, and gamma with local computation and cross-area integration (NHA Health, 2023; Muthukumaraswamy, 2013). The white matter tracts connecting these cortical regions are not only the physical medium for this communication but also the main area affected in mTBI. Therefore, disruptions in oscillatory coordination are among the most sensitive indicators of brain injury.

Essentially, this oscillatory coordination is not a fixed property of the brain but fluctuates continuously over time, resulting in moment-to-moment changes in cognitive state and network organization (Hutchison et al., 2013). The temporal pattern of these fluctuations, meaning how often the brain transitions between different connectivity states and how long it dwells in each, is what dynamic functional connectivity analysis is about, and what turns out to be disrupted in conditions like mTBI (Antonakakis et al., 2019, 2017).

2.2 Mild Traumatic Brain Injury

Mild traumatic brain injury (mTBI) accounts for approximately 90% of all traumatic brain injuries (Len and Neary, 2011). It is caused when a biomechanical force is applied to the head, resulting in a brief alteration of consciousness, post-traumatic amnesia of less than 24 hours, and a Glasgow Coma Scale score between 13 and 15 (Kay et al., 1993; Assistant Secretary, Department of Defense, 2007). Although it is considered mild, around 20% of patients develop persistent post-concussion symptoms — headaches, fatigue, sleep disturbance, and difficulties with attention, memory, and processing speed — that can last months after the initial event and even lead to long-term impairment (Sigurdardottir et al., 2009; Vanderploeg et al., 2005; De Monte et al., 2006).

The underlying mechanism is diffuse axonal injury: the acceleration-deceleration forces of the impact stretch and damage long-range white matter tracts that connect cortical regions across the brain (Huang et al., 2012; Zhang et al., 2020). These tracts physically sustain coordinated oscillatory activity between brain areas, so their disruption causes direct changes in phase synchronization patterns (Antonakakis et al., 2017). Because this damage is microscopic and spatially distributed rather than focal, CT and MRI scans tend to appear normal (Kirkwood et al., 2006; Bigler and Orrison, 2004). Conventional

structural neuroimaging simply lacks the sensitivity to detect subtle alterations in brain dynamics (Eierud et al., 2014).

However, functional neuroimaging with sufficient temporal resolution can detect such mild alterations: disrupted functional interdependencies between cortical areas, reduced phase synchronization between brain regions, lower global network efficiency, and decreased flexibility in transitions between connectivity states compared to healthy individuals (Omidvarnia et al., 2025; Antonakakis et al., 2019, 2020). These changes in the temporal dynamics of functional connectivity are precisely what chronnectomic features such as transition probability and mean dwell time manage to capture, and why they have consistently emerged as discriminative biomarkers of mTBI.

2.3 Magnetoencephalography

Magnetoencephalography (MEG) is a non-invasive functional neuroimaging technique that measures the weak magnetic fields generated by the synchronous electrical activity of large populations of cortical neurons. When thousands of pyramidal neurons fire in synchrony with aligned orientations, the resulting primary currents produce magnetic fields on the order of 10^{-13} Tesla that penetrate the skull and scalp without distortion (Hämäläinen et al., 1993). MEG involves measuring such magnetic fields using arrays of superconducting quantum interference devices (SQUIDs) arranged in a helmet-shaped scanner surrounding the subject’s head (MRC CBU Imaging, 2023).

MEG’s principal advantage for this work is its combination of high temporal resolution, since it captures neural dynamics at the millisecond timescale, with reasonable spatial localization. Unlike EEG, MEG signals are not distorted by the differential conductivity of skull and scalp tissue. Also, by measuring neural activity directly rather than the indirect hemodynamic responses captured by fMRI, MEG can resolve with precision the rapid, frequency-specific oscillatory dynamics that characteristically underlie functional connectivity (Ahlfors and Mody, 2019).

Only when large populations of neurons fire simultaneously does a sufficiently strong magnetic field emerge to be detectable. The resulting activity spans several frequency bands — delta (δ , 0.5–4 Hz), theta (θ , 4–8 Hz), alpha (α , 8–13 Hz), beta (β , 13–30 Hz), and gamma (γ , above 30 Hz) — each reflecting distinct brain states and cognitive processes, with amplitudes and dominant frequencies shifting across wakefulness, sleep, and development (Sanei and Chambers, 2007; Velmurugan et al., 2014). These properties have made MEG an increasingly valuable tool in clinical neuroscience. Where structural

imaging consistently fails to detect the subtle functional alterations caused by conditions like mTBI, MEG has proven sensitive to disrupted connectivity patterns (Dimitriadis et al., 2015; Antonakakis et al., 2016), aberrant slow-wave activity (Huang et al., 2012), and network-level abnormalities across a range of neurological conditions including mild cognitive impairment (Maestú et al., 2015), multiple sclerosis (Tewarie et al., 2015), and reading difficulties (Dimitriadis et al., 2016).

2.4 Functional Connectivity and Its Dynamic Analysis

Functional connectivity (FC) refers to the statistical dependence between the activity of spatially distinct brain areas, capturing the degree to which these areas oscillate in a coordinated way over time (Friston, 2011). Rather than describing fixed anatomical connections as structural connectivity does, FC is inherently dynamic. This means it reflects the brain’s real-time reorganization in response to internal states and external inputs. In practice, it is represented as a graph where nodes denote brain regions, while edges denote the strength of their pairwise synchronization. Figure 2.2 provides a visual example of such a representation, with warmer colors indicating stronger connections.

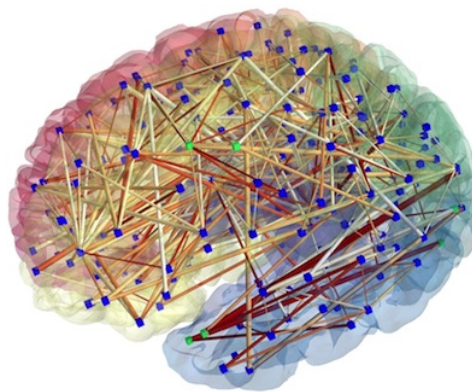


Figure 2.2: A functional connectivity graph mapped onto the cortical surface. Adapted from European Society of Radiology (2014).

Synchronization strength is assessed with measures such as phase locking, amplitude correlation, or coherence (Allen et al., 2012). Based on this graph-based approach, researchers can apply network science to brain data and yield metrics such as global efficiency, clustering coefficient, and characteristic path length that explain exactly how the information flows across the brain (Rubinov and Sporns, 2010).

According to classical models, FC is considered stationary, meaning that a single connectivity matrix is computed over the whole recording period. However, as Ahrends and Vidaurre (2023) note, static FC captures only instantaneous relationships — if the time points within a session were shuffled, the static FC estimate would not change. In other words, it is blind to the temporal dynamics of how connectivity evolves, which is precisely where clinically important information often resides. Dynamic functional connectivity (dFC) analysis addresses this issue by estimating connectivity within a series of short sliding windows. This way, a time-varying sequence of connectivity graphs is produced that captures how the brain’s network organization fluctuates over seconds (Hutchison et al., 2013; Antonakakis et al., 2019).

2.5 Network Microstates and Chronnectomic Features

The time-varying sequence of dFC graphs is characterized by complexity, but also high dimensionality. Therefore, a more compact and clinically interpretable representation can be obtained with network microstate analysis. The sequence can be compressed into a small number of discrete brain states, so that a symbolic time series is produced. This time series describes how the brain transitions between states over time.

The theoretical basis for this reduction comes from spectral graph theory: the eigen-spectrum of a graph’s normalized Laplacian represents the network’s synchronization structure, capturing properties such as how efficiently information can propagate across the graph (Chung and Graham, 1997). For example, smaller eigenvalues indicate more globally integrated connectivity patterns. This provides a principled way to represent each connectivity snapshot as a low-dimensional vector without losing essential topological information.

These spectral vectors are then grouped into prototypical states using vector quantization. This entails several methods that find a small set of codebook vectors in order to minimize the reconstruction error between the original sequence and its symbolic approximation (Martinetz et al., 1993). Some of these methods are k-means clustering, Independent Component Analysis, and Hidden Markov Models, each with different assumptions about the underlying dynamics (Dimitriadis et al., 2019). The choice of quantization algorithm and the number of states k are methodological decisions fully described later in this thesis. The output is a symbolic time series of state assignments; a compact depiction of how the brain changes connectivity patterns over time.

From this symbolic time series, a set of temporal and graph-theoretic features, also known as chronnectomics (Allen et al., 2012), can be extracted to characterize each subject’s brain network dynamics. These include measures of how frequently the brain transitions between states (transition probability), how long it spends in each state (occupancy time and dwell time), and also graph-theoretic properties of the connectivity structure corresponding to each state, such as global efficiency and clustering coefficient (Antonakakis et al., 2020; Rubinov and Sporns, 2010). Together they provide a complete, interpretable description of the brain’s dynamic organization that has been proven to carry discriminative clinical information in mTBI (Antonakakis et al., 2019, 2020). The specific features extracted in this thesis and the justification for their selection are described in the Methodology chapter.

2.6 Kolmogorov-Arnold Networks (KANs)

2.6.1 The Kolmogorov-Arnold Representation Theorem

The theoretical foundation of KANs rests on the Kolmogorov-Arnold Representation Theorem, established independently by Andrey Kolmogorov (Kolmogorov, 1957) and Vladimir Arnold (Arnold, 1957). The theorem states that any multivariate continuous function f defined on a bounded domain can be exactly represented as a finite composition of continuous univariate functions and the binary operation of addition. Formally, for a smooth function $f : [0, 1]^n \rightarrow \mathbb{R}$:

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \varphi_{q,p}(x_p) \right)$$

where $\varphi_{q,p} : [0, 1] \rightarrow \mathbb{R}$ are inner univariate functions and $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$ are outer univariate functions. The theorem implies that the only true multivariate operation is addition: every other function decomposes into sums of univariate ones.

For machine learning, this means that learning a high-dimensional function can theoretically be reduced to learning a collection of one-dimensional functions, which are much easier to approximate and interpret. Despite its mathematical elegance, the theorem was historically considered impractical for this purpose, since the inner functions it guarantees may be non-smooth or even fractal in the worst case (Poggio et al., 2020). Liu et al. (2024) resolved this by noting that functions in scientific and biomedical domains are typically smooth in practice, making B-spline approximation of the inner functions both

reliable and tractable. Motivated by this insight, they generalized the initial two-layer representation to arbitrary width and depth, parametrizing each univariate function as a learnable B-spline. The resulting architecture — Kolmogorov-Arnold Networks (KANs) — was accepted at ICLR 2025 and demonstrated competitive or superior accuracy to MLPs on scientific tasks, alongside significantly better interpretability (Liu et al., 2024).

2.6.2 KAN Architecture

KANs generalize the original two-layer Kolmogorov-Arnold representation to networks of arbitrary depth and width. The shape of a KAN is defined by an integer array $[n_0, n_1, \dots, n_L]$, where n_i is the number of nodes at layer i . A KAN layer with n_{in} inputs and n_{out} outputs is defined as a matrix of learnable univariate functions (Liu et al., 2024):

$$\Phi = \{\varphi_{q,p}\}, \quad p = 1, \dots, n_{\text{in}}, \quad q = 1, \dots, n_{\text{out}}$$

where each function $\varphi_{q,p}$ has trainable parameters and is applied to a single scalar input. The activation value of node $(l+1, j)$ is then the sum of all incoming post-activations:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \varphi_{l,i,j}(x_{l,i}), \quad j = 1, \dots, n_{l+1}$$

A complete KAN of depth L is the composition of L such layers:

$$\text{KAN}(\mathbf{x}) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_0)(\mathbf{x})$$

The main difference in architecture compared to MLPs is that in MLPs, fixed nonlinearities are applied at the nodes, and learnable scalar weights sit on the edges. In KANs, on the other hand, learnable nonlinear functions are placed on the edges themselves, and nodes perform only summation (Liu et al., 2024; Goodfellow et al., 2016). This means that every parameter in the network corresponds to a piece of a visualizable function rather than an anonymous scalar weight.

Figure 2.3 illustrates both architectures side by side for a network with two inputs, a hidden layer of three nodes, and a single output, making the structural differences immediately visible.

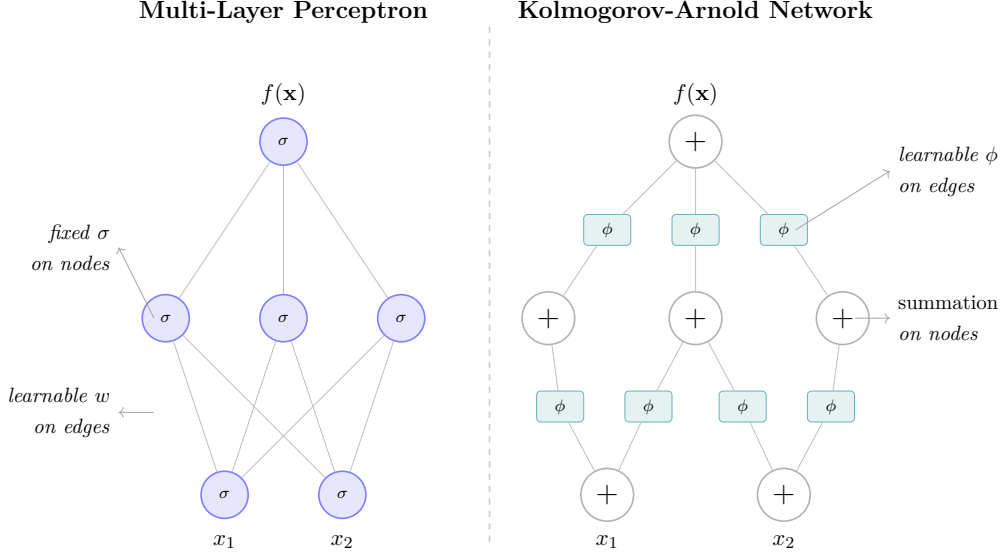


Figure 2.3: Architectural comparison between MLP and KAN.

2.6.3 B-Spline Parametrization

Each univariate activation function on a KAN edge is parametrized as the sum of a fixed residual basis function and a learnable spline (Liu et al., 2024):

$$\varphi(x) = w \cdot (b(x) + \text{spline}(x))$$

where $b(x) = \text{SiLU}(x) = x/(1 + e^{-x})$ is the residual basis function that stabilizes optimization by providing smooth default behaviour before any training occurs (Elfwing et al., 2018), and:

$$\text{spline}(x) = \sum_i c_i \cdot B_i(x)$$

where $B_i(x)$ are B-spline basis functions of order k defined over G grid intervals with trainable coefficients c_i . The scalar w is initialized by Xavier initialization to control the overall activation magnitude.

B-splines are piecewise polynomial functions where each piece is a low-degree polynomial and adjacent pieces join smoothly at the knot boundaries (de Boor, 1978). Essentially, a B-spline is a linear combination of basis functions, which are non-negative and locally supported. This means that each basis function is non-zero only over a small portion of the input domain, enabling precise local adaptation without disturbing behaviour elsewhere

(Chaudhuri, 2021). Another practical advantage is grid extensibility: the model can be made progressively more accurate by refining the spline grid without retraining from scratch (Liu et al., 2024).

Compared to MLPs with ReLU activations, KANs with B-splines exhibit a substantially faster neural scaling law. Liu et al. (2024) show that test error theoretically decreases as $\ell \propto N^{-4}$ with the number of model parameters N for cubic splines, compared to a scaling exponent of approximately 1 for standard MLPs. This superior parameter efficiency is particularly advantageous for small biomedical datasets, where the risk of overfitting is high.

2.6.4 Sparsification and Feature Importance

A key property of KANs is their built-in mechanism for identifying the most important input features (Liu et al., 2024). During training, a sparsification regularization term is added to the prediction loss. This encourages the model to focus on a small number of highly active edges, effectively pruning irrelevant connections. The total training objective is:

$$\ell_{\text{total}} = \ell_{\text{pred}} + \lambda \left(\mu_1 \sum_l |\Phi_l|_1 + \mu_2 \sum_l S(\Phi_l) \right)$$

where ℓ_{pred} is the prediction loss, λ controls the overall regularization strength, and μ_1 and μ_2 are weighting coefficients. The first term represents L1 regularization on the activation functions, defined as the mean absolute magnitude over training inputs:

$$|\varphi|_1 = \frac{1}{N_p} \sum_{s=1}^{N_p} |\varphi(x^{(s)})|$$

The second term represents an entropy penalty that encourages a non-uniform distribution of activation magnitudes among edges:

$$S(\Phi) = - \sum_i \sum_j \frac{|\varphi_{i,j}|_1}{|\Phi|_1} \cdot \log \left(\frac{|\varphi_{i,j}|_1}{|\Phi|_1} \right)$$

This way, the network tends to favor solutions where a sparse set of edges carries the dominant information. As noted by Liu et al. (2024), the entropy term is essential: without it, L1 regularization alone produces multiple identical low-magnitude connections

rather than an actually sparse architecture. Together, they lead the network toward a solution where only a few edges are active — precisely what is needed for a clear biomedical interpretation.

The importance score of an input feature is calculated as the sum of the L1 norms of all activation functions connected to that input. This method is completely different from external attribution methods such as SHAP (Lundberg and Lee, 2017) or permutation importance (Breiman, 2001), which include separate calculations after the model’s training. In a KAN, however, the model’s own learned representation is the feature importance measure.

Chapter 3 – Prior Work

Several previous studies have been used as the foundation for this thesis and the most important ones are presented in this Chapter. Figure 3.1 summarises the full pipeline, from raw MEG recordings to the thresholded dominant dynamic functional connectivity graphs that serve as the starting point for this work.

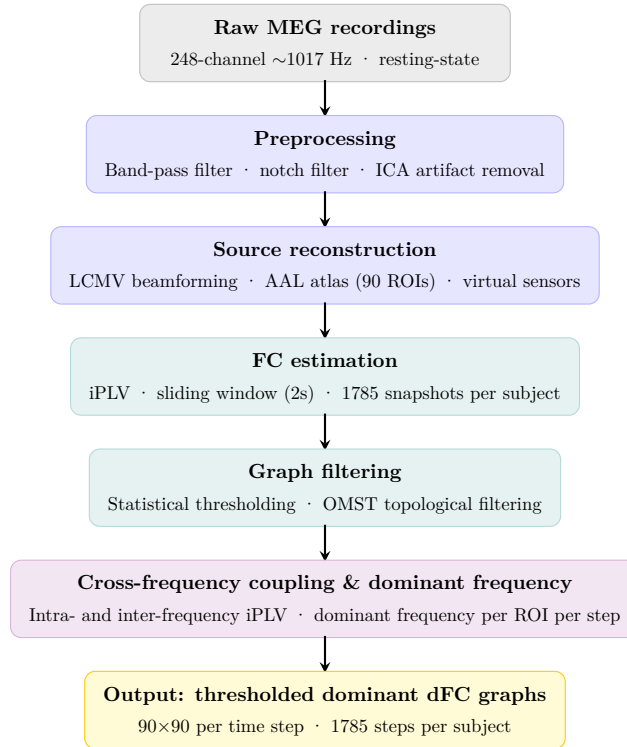


Figure 3.1: The preprocessing and connectivity extraction pipeline established by Antonakakis et al. Antonakakis et al. (2020) and Dimitriadis et al. Dimitriadis et al. (2017a).

3.1 MEG Preprocessing and Source Reconstruction

The resting-state MEG dataset evaluated in this thesis was originally acquired and preprocessed using the source-reconstruction pipeline established by Antonakakis et al. (2020) in their previous studies on mTBI classification. The experimental recordings were collected with a 248-channel whole-head axial gradiometer system at a sampling frequency of approximately 1,017 Hz, with participants in a supine position, under eyes-closed resting conditions. From the initial ten-minute recording sessions, approximately five minutes of

artifact-free, continuous data segments were isolated for evaluation (Antonakakis et al., 2019, 2020).

The raw data cleaning protocol involved a sequence of automated and manual inspection steps (Antonakakis et al., 2017). First, compromised channels were identified and corrected via spherical spline interpolation. After that, a 0.1–100 Hz band-pass filter was applied to the continuous signals, as well as a notch filter at 60 Hz to remove power line noise. To eliminate electrophysiological artifacts, Independent Component Analysis (ICA) decomposition routines were used within the FieldTrip environment (Oostenveld et al., 2011). Ocular and cardiac components were flagged and dropped by thresholding their empirical kurtosis, skewness, and Rényi entropy values (Antonakakis et al., 2017).

Source reconstruction used atlas-based Linearly Constrained Minimum Variance (LCMV) beamforming (Van Veen et al., 1997). Sensor geometries were co-registered to a standard T1-weighted MRI template provided by SPM8 (Weiskopf et al., 2011), and the brain volume was parcellated into 90 cortical and subcortical regions of interest (ROIs) according to the Automated Anatomical Labeling (AAL) atlas (Tzourio-Mazoyer et al., 2002). A realistic spherical head model with 5,061 sources at 6 mm grid resolution defined the forward problem (Nolte et al., 2004). Instead of relying on a simple spatial average within each ROI, virtual sensors were derived by weighting voxel time series by their normalized node strength within an intra-ROI connectivity graph (Dimitriadis et al., 2018). This produced a single, high-SNR, and functionally informed time series per region. Ultimately, the result was 90 source-level time series per subject, which constituted the baseline input for this thesis.

3.2 Dynamic Functional Connectivity Graph Extraction and Filtering

The construction and filtering of dynamic functional connectivity graphs (DFCGs) were thoroughly described in the methodology of Antonakakis et al. (2020) and in a series of related papers by Dimitriadis et al. (2017a,b). To capture the localized, time-varying spectral properties of the source-reconstructed data, the 90 virtual sensor time series were first decomposed into standard canonical frequency bands: delta (δ : 0.5–4 Hz), theta (θ : 4–8 Hz), alpha 1 (α_1 : 8–10 Hz), alpha 2 (α_2 : 10–13 Hz), beta 1 (β_1 : 13–20 Hz), beta 2 (β_2 : 20–30 Hz), gamma 1 (γ_1 : 30–45 Hz), and gamma 2 (γ_2 : 55–90 Hz).

Network interactions were quantified using the imaginary part of the Phase Locking Value (iPLV) (Palva et al., 2018). This specific connectivity estimator was selected because

it ignores zero-lag coupling, effectively bypassing the artificial correlations caused by spatial volume conduction (Nolte et al., 2004). Phase synchronization was estimated within a sliding 2-second window that shifted forward by just one sample at a time across the five-minute recording (Antonakakis et al., 2020). This high-density temporal sampling generated a sequence of 1,785 discrete connectivity snapshots per subject for each frequency band, with each snapshot saved as a 90×90 connectivity matrix.

Before building cross-frequency models, these raw matrices underwent a two-stage filtering process to remove random background noise. First, statistical filtering was performed using surrogate analysis: 10,000 phase-scrambled surrogate time series were generated per window, and any connection that failed an FDR-corrected significance test ($p < 0.05$, $q = 0.01$) was zeroed out (Aru et al., 2015; Benjamini and Hochberg, 1995). Second, topological filtering using the Orthogonal Minimal Spanning Tree (OMST) was applied (Dimitriadis et al., 2017a,b). The OMST technique iteratively extracted orthogonal minimum spanning trees to isolate the most functionally dominant pathways while maximizing global efficiency at a low connection cost (Dimitriadis et al., 2017b). These two steps removed spurious connections while retaining the core functional architecture at each time step (Antonakakis et al., 2020). The resulting filtered tensors — $1,785 \times 90 \times 90$ per subject per frequency band — served as the input to the next step.

3.3 Cross-Frequency Coupling and Dominant Frequency Identification

The connectivity framework from Dimitriadis et al. (2017a,b) tracked both intra-frequency and cross-frequency interactions to map the full complexity of the connectome. Intra-frequency coupling measured phase synchronization between regions within the same spectral band, estimated via iPLV across all the standard bands (δ , θ , α , β , γ) (Antonakakis et al., 2020). Cross-frequency coupling, on the other hand, captured phase-to-amplitude coupling (PAC). PAC tracked the mechanism where the phase of a slow, global lower-frequency rhythm modulated the amplitude envelope of a faster, localized higher-frequency oscillation (Jensen and Colgin, 2007). This hierarchical gating coordinated large-scale distributed brain networks, a mechanism that was often disrupted or desynchronized by white matter microstructural damage in mTBI (Huang et al., 2009). Antonakakis et al. (2016) proved the clinical value of this feature, showing that cross-frequency PAC metrics, derived via mutual information across six distinct frequency pairs, distinguished mTBI patients from healthy controls with high accuracy. Their work noted that healthy subjects consistently maintained denser, more integrated, and resilient

cross-frequency networks (Antonakakis et al., 2016).

A critical step from the Dimitriadis et al. (2017a) methodology was the identification of the dominant frequency for each brain region. This meant determining the oscillatory band — intra- or inter-frequency — that demonstrated the highest synchronization and most strongly drove the functional connectivity of a given node at any precise time step (Dimitriadis et al., 2017a). So, rather than tracking all eight frequency bands independently, this protocol collapsed the multi-layered tensor by dynamically isolating the specific frequency channel that carried the most prominent information for each AAL region at each moment in time (Antonakakis et al., 2019).

The practical output of this cross-frequency integration was a thresholded dominant functional connectivity graph of dimensionality 90×90 per time step, constructed from iPLV values at the dominant frequency per region and filtered via OMST (Dimitriadis et al., 2017a,b). All this time-varying network information gathered into a single graph per time step sharpened group-level clinical differences among subjects (Antonakakis et al., 2019, 2020) and was one of the features that made the present pipeline stand out from simpler single-band analyses.

For the scope of this thesis, all preceding preprocessing, source reconstruction, and multi-frequency filtering steps were applied in these prior studies. The resulting time-varying thresholded dominant connectivity graphs were provided as the finalized starting dataset. This dataset was the input to the microstate extraction stage described in the Methodology chapter that follows, where the normalized Laplacian eigenspectrum, Neural Gas vector quantization, chronnectomic feature extraction, and KAN-based classification are presented in full detail.

Chapter 4 – Methodology

This chapter describes the methodological steps of the proposed pipeline, which is summarised in Figure 4.1. Each component is fully detailed in the sections that follow.

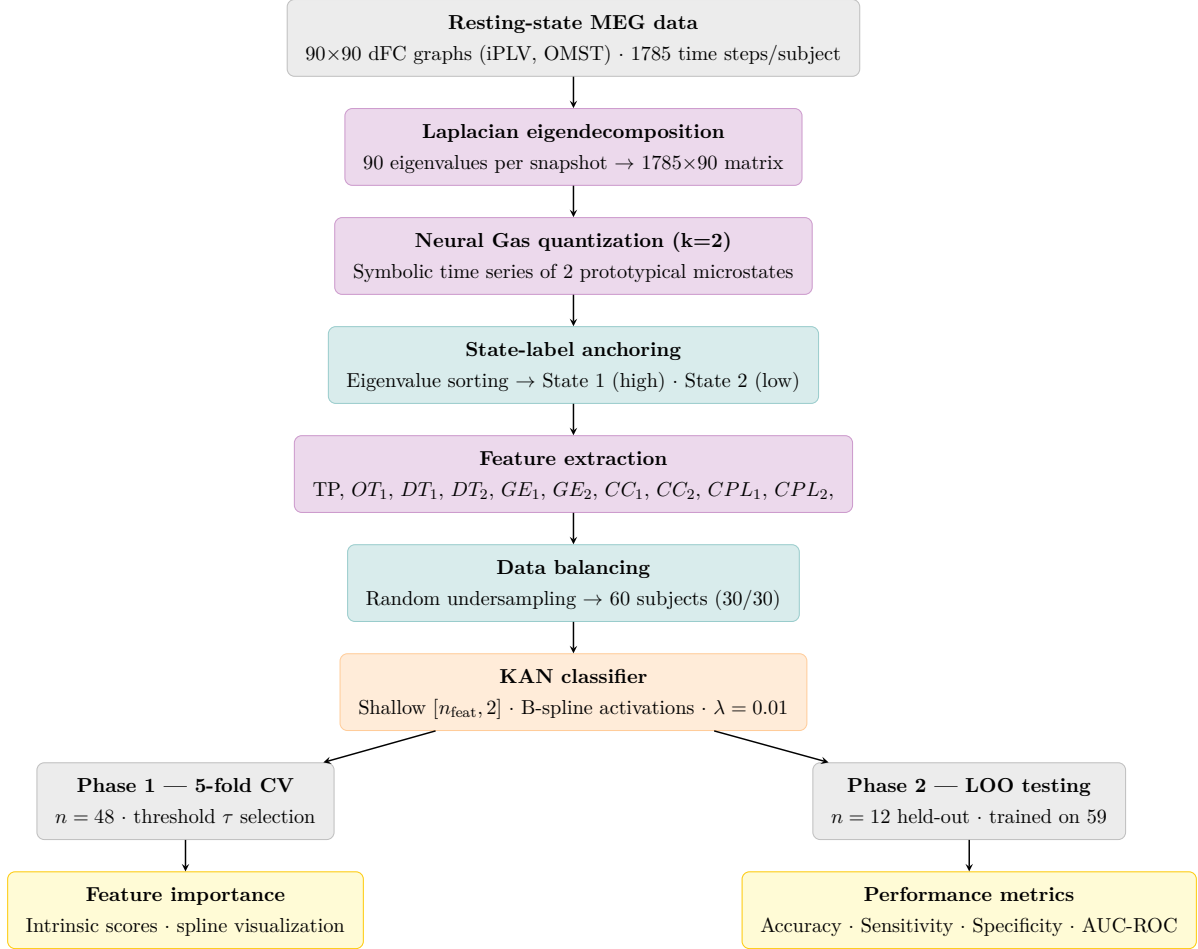


Figure 4.1: End-to-end pipeline for mTBI classification from resting-state MEG data.

4.1 Laplacian Eigendecomposition

The starting point of the analysis of this thesis is the set of thresholded dominant functional connectivity graphs produced by the pipeline described in Chapter 3. Each one of the 78 subjects of the dataset is represented by a sequence of 1,785 connectivity snapshots. Each snapshot includes a 90×90 weighted adjacency matrix G_t capturing the dominant phase synchronization structure of the brain at time step t .

Instead of using these high-dimensional matrices as direct input, each one of them was compressed into a compact spectral representation. This means that for each snapshot, the normalized Laplacian of the graph was computed as (Chung and Graham, 1997):

$$L = I - D^{-1/2} \cdot G \cdot D^{-1/2}$$

where I is the identity matrix, G is the 90×90 thresholded dominant connectivity graph, and D is the diagonal degree matrix with $D_{ii} = \sum_j G_{ij}$. Isolated nodes with zero degree produced NaN entries, which were set to zero before proceeding. Eigenvalue decomposition of L then yielded 90 real-valued eigenvalues per snapshot, sorted in ascending order. Intuitively, smaller eigenvalues indicate more globally integrated connectivity patterns, while larger ones reflect more locally segregated structure (Chung and Graham, 1997). The full sequence of eigenvalue vectors across all 1,785 time steps formed a 1785×90 matrix per subject, which served as the input to the following stage.

4.2 Neural Gas Quantization and State-Label Consistency

4.2.1 Neural Gas Quantization

The sequence of spectral vectors was quantized into $k = 2$ prototypical brain states using the Neural Gas algorithm (Martinetz et al., 1993). Two states were chosen following Antonakakis et al. (2020), who showed that at $k = 2$ the distortion error between the original eigenvalue time series and the reconstructed symbolic approximation remained below 2% for every subject. This essentially meant that two prototypes were sufficient to capture dominant connectivity dynamics without meaningful information loss.

So, the algorithm initialized two prototype vectors that were randomly selected from the input data. It then performed a series of stochastic gradient descent updates over the spectral vectors, also presented in randomized order. At each step, both prototypes were adapted toward the current input, with a learning rate scaled by each prototype’s rank in the distance ordering. Both learning rate and neighbourhood width decayed exponentially over training, leading to a coarse-to-fine convergence. Unlike the k-means algorithm, which updates only the nearest prototype, Neural Gas applies rank-weighted updates to all prototypes at the same time, making convergence substantially less sensitive to initialization (Martinetz et al., 1993). The output was two codebook vectors representing

the dominant low- and high-connectivity patterns, and a symbolic time series of length 1,785 assigning each snapshot to its nearest prototype.

4.2.2 State-Label Consistency

One of the main challenges in group-level microstate analysis is state-label consistency. Because the Neural Gas algorithm runs independently per subject, the label assigned to each prototype is mathematically arbitrary. Without a principled anchoring strategy, “State 1” in one subject may correspond to a high-connectivity topology while in another subject it corresponds to a low-connectivity topology. Consequently, group-level comparisons of state-specific features can be unreliable.

Antonakakis et al. (2020) addressed this by computing pairwise cosine similarity between prototypes across subjects and applying K-means clustering to align them at the group level. In this thesis, a deterministic alternative was adopted: for each subject, the mean of all 90 eigenvalues was computed separately for each prototype, and the prototype exhibiting the lower mean was assigned the label “State 1” while the prototype with the higher mean was assigned “State 2”. Given that lower eigenvalues of the normalized Laplacian correspond to more globally integrated, higher-connectivity network configurations (Chung and Graham, 1997), this sorting criterion ensures that State 1 and State 2 carry a consistent neurophysiological interpretation across all subjects; State 1 corresponds to the higher-connectivity microstate and State 2 to the lower-connectivity one.

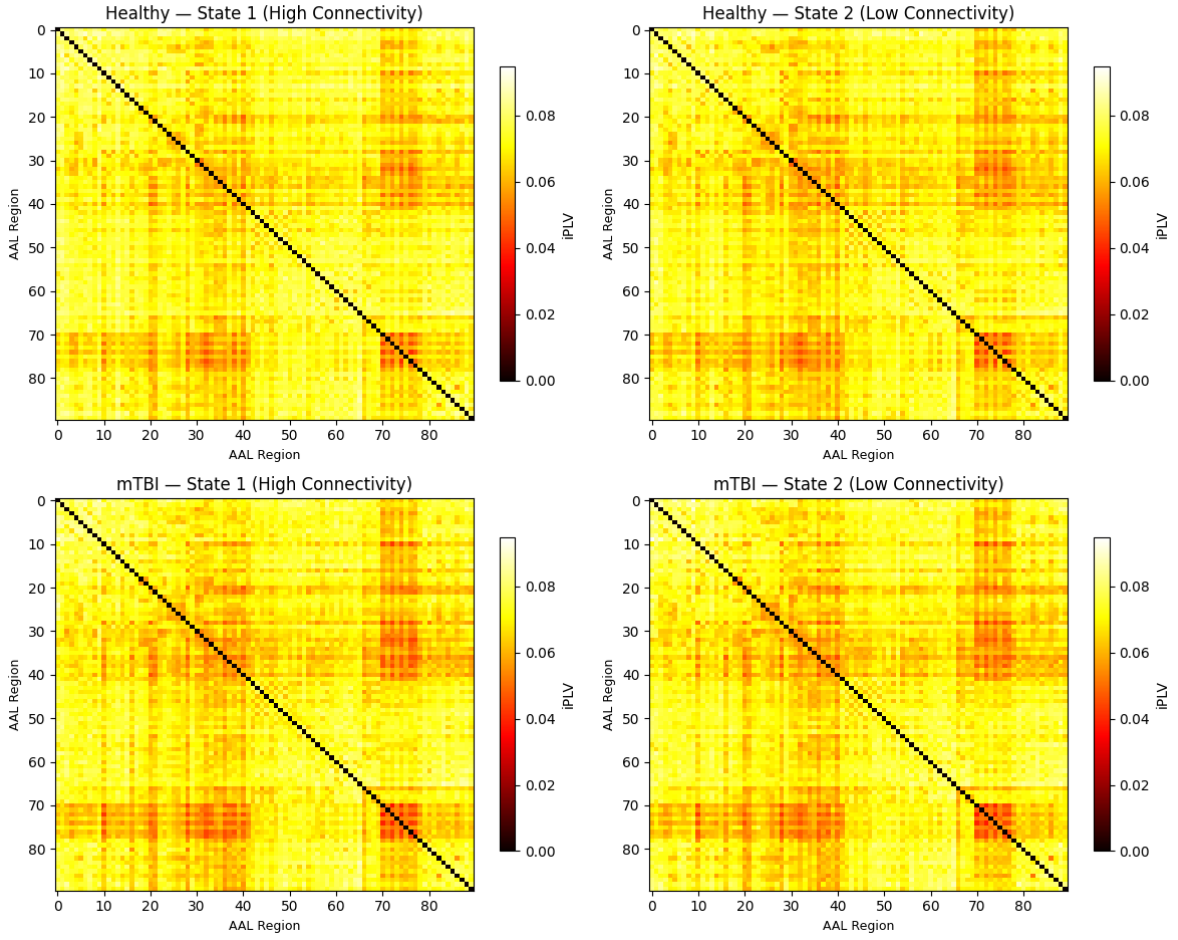


Figure 4.2: Mean connectivity matrices (iPLV) for each group and microstate.

Figure 4.2 shows the group-level mean connectivity matrices for each microstate and group. The mTBI matrices appear slightly warmer (lower iPLV values) compared to healthy subjects across both states, pointing to a mild but consistent reduction in phase synchronization strength. This aligns with the fact that damaged white matter tracts weaken long-range oscillatory coordination between cortical regions (Antonakakis et al., 2016; Huang et al., 2012). However, the difference is subtle and spatially distributed rather than focal, so it is not sufficient on its own to distinguish the two groups. This is exactly what motivates the dynamic microstate analysis that follows.

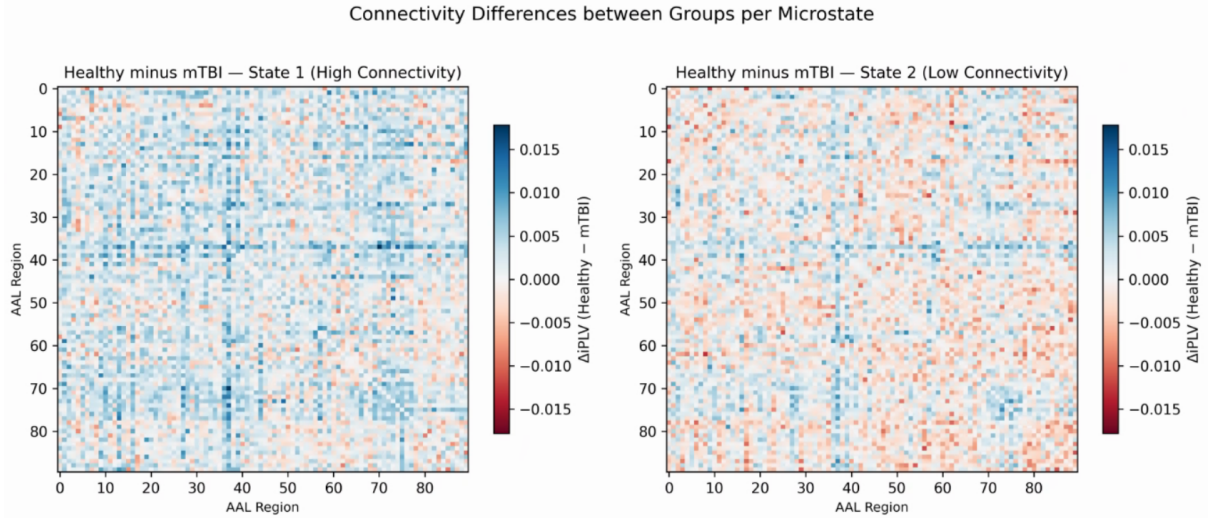


Figure 4.3: Difference matrices (healthy minus mTBI) per microstate.

Figure 4.3 shows the difference matrices (healthy minus mTBI) per state, making connectivity differences between groups easier to detect visually. In State 1, blue predominates over red, indicating that healthy subjects consistently exhibit stronger connectivity than mTBI patients across most region pairs. This is particularly visible around regions 35–40 and 70–80, corresponding to the right temporal cortex and subcortical areas such as the thalamus and basal ganglia, respectively. A small number of red patches are also present, where mTBI patients show slightly stronger local connectivity, consistent with the focal hyperconnectivity findings reported in Antonakakis et al. (2016). In State 2, the pattern is more spatially diffuse and mixed with no clear dominant direction, suggesting the two groups are more similar during that state. Overall, group differences appear more pronounced during the high-connectivity microstate, which motivates keeping State 1-specific and State 2-specific features separate in the classification pipeline, rather than collapsing across states.

4.3 Feature Extraction

From the label-consistent symbolic time series, a set of temporal and graph-theoretic features was extracted per subject. These features constituted the input to the classification model.

4.3.1 Temporal Chronnectomic Features

Three temporal features were extracted from the symbolic time series, following the chronnectomic framework established by Antonakakis et al. (2020):

- **Transition Probability (TP):** The fraction of consecutive time windows in which a state change occurred. It quantifies how dynamically the brain switches between the two connectivity regimes, with higher values reflecting greater temporal flexibility. It was computed as:

$$\text{TP} = \frac{\sum_{t=1}^{T-1} \mathbf{1}[c_t \neq c_{t+1}]}{T - 1}$$

where c_t denotes the state label at time step t and $T = 1,785$.

- **Occupancy Time (OT):** The proportion of the total recording time spent in State 1 (higher connectivity). It captures whether one connectivity regime dominated the session, with lower values indicating that the brain spent more time in the higher-connectivity State 2. It was computed as:

$$\text{OT} = \frac{\sum_{t=1}^T \mathbf{1}[c_t = 1]}{T}$$

- **Dwell Time (DT):** The mean duration of consecutive periods spent within each state before transitioning, computed separately for State 1 and State 2. It complements occupancy by revealing whether the brain visited a state in many brief episodes or fewer sustained ones. For state $s \in \{1, 2\}$, let $r_1, r_2, \dots, r_m$ denote the lengths of all uninterrupted runs of label s in the symbolic time series. The dwell time is then:

$$\text{DT}_s = \frac{1}{m} \sum_{i=1}^m r_i$$

4.3.2 Graph-Theoretic Features

For each microstate, the mean connectivity graph was computed by averaging the raw 90×90 connectivity matrices of all time steps assigned to that state. Three graph-theoretic metrics were then extracted from each mean graph using the Brain Connectivity Toolbox (Rubinov and Sporns, 2010):

- **Global Efficiency (GE):** The average inverse shortest path length across all node

pairs, measuring the capacity for parallel information transfer across the network:

$$GE = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}}$$

where N is the number of nodes and d_{ij} is the shortest weighted path length between nodes i and j . Computed using the weighted formulation `efficiency_wei` from the Brain Connectivity Toolbox.

- **Clustering Coefficient (CC):** The mean clustering coefficient across all nodes, measuring the tendency for local network segregation:

$$CC = \frac{1}{N} \sum_{i=1}^N \frac{\sum_{j,h} (w_{ij}w_{ih}w_{jh})^{1/3}}{k_i(k_i-1)}$$

where w_{ij} are the edge weights normalized by the maximum weight, and k_i is the degree of node i . Computed using `clustering_coef_wu` from the Brain Connectivity Toolbox.

- **Characteristic Path Length (CPL):** The average shortest path length between all pairs of nodes, measuring the overall routing efficiency of the network:

$$CPL = \frac{1}{N} \sum_{i=1}^N \left(\frac{\sum_{j \neq i} d_{ij}}{N-1} \right)$$

where d_{ij} is the shortest weighted path length between nodes i and j . A weighted distance matrix was first derived from the connectivity graph using `distance_wei`, which converts edge weights to lengths by taking their inverse. From this distance matrix, the characteristic path length was computed using `charpath` from the Brain Connectivity Toolbox.

These metrics were extracted separately for State 1 and State 2, yielding six additional features: GE_1 , CC_1 , CPL_1 , GE_2 , CC_2 , CPL_2 .

4.3.3 Full Feature Vector

The full feature vector per subject consisted of ten features:

$$\mathbf{x} = [TP, OT_1, DT_1, DT_2, GE_1, CC_1, CPL_1, GE_2, CC_2, CPL_2]$$

All features were standardized to zero mean and unit variance using `StandardScaler`

from `scikit-learn` (Pedregosa et al., 2011), fitted exclusively on the training data at each cross-validation fold to prevent data leakage. This ten-feature vector served as the full input to the classification pipeline, with a subsequent feature selection step applied within each cross-validation fold to identify the most informative subset for the final classifier.

4.4 Data Balancing

The dataset consisted of 49 healthy controls and 30 mTBI patients, presenting a class imbalance that could bias classifier training toward the majority class (Chen et al., 2024). To address this, a random subsampling method was used, which included 100 independent experimental runs to ensure statistical robustness. At each iteration, 30 healthy subjects were randomly selected from the 49 available, producing a balanced dataset of 60 subjects (30 healthy, 30 mTBI). The random seed was set to the run index to make sure that each run selected a different subset of healthy subjects. This introduced controlled variability across runs while keeping the code fully reproducible, since the same sequence of subsets was generated every time the experiment was executed. This strategy was preferred over oversampling methods such as SMOTE (Chawla et al., 2002), as synthetic data generation is often unreliable with small sample sizes. So, undersampling from the pool of 49 healthy individuals was chosen to preserve the group’s natural data distribution (Chen et al., 2024).

4.5 KAN Architecture

A shallow Kolmogorov-Arnold Network of shape $[n_{\text{feat}}, 2]$ was used as the binary classifier, where n_{feat} is the number of features selected by the feature importance analysis (between 2 and 10 from the full 10-dimensional feature vector). No hidden layer was used for two reasons. First, with only 60 subjects after balancing, a deeper architecture would introduce far more parameters than the dataset can reliably support, increasing the risk of overfitting. Second, a single-layer KAN reduces the classification decision to a direct composition of $n_{\text{feat}} \times 2$ univariate spline functions — one per feature-output pair — each of which is independently visualizable and neurophysiologically interpretable (Liu et al., 2024).

The KAN was implemented using the `pykan` library (Liu et al., 2024) with cubic B-splines ($k = 3$) and a grid of $G = 3$ intervals, yielding $G + k = 6$ basis functions per activation

function. These settings follow the defaults of Liu et al. (2024) and were chosen to keep the number of learnable parameters per activation function small relative to the available training data. As a result, the risk of overfitting is reduced, while retaining sufficient flexibility to capture nonlinear relationships between features and class labels.

The cross-entropy loss function was used as the prediction loss ℓ_{pred} . Cross-entropy measures the divergence between the predicted class probabilities and the true labels, penalizing confident wrong predictions more heavily than uncertain ones (Goodfellow et al., 2016). It is the standard choice for binary classification tasks in which the model outputs a softmax probability, as is the case here.

Figure 4.4 shows the full $[10, 2]$ architecture schematically, representing the case in which all ten features are used as inputs. As discussed in the following section, the number of active input nodes may vary depending on which features are selected during training.

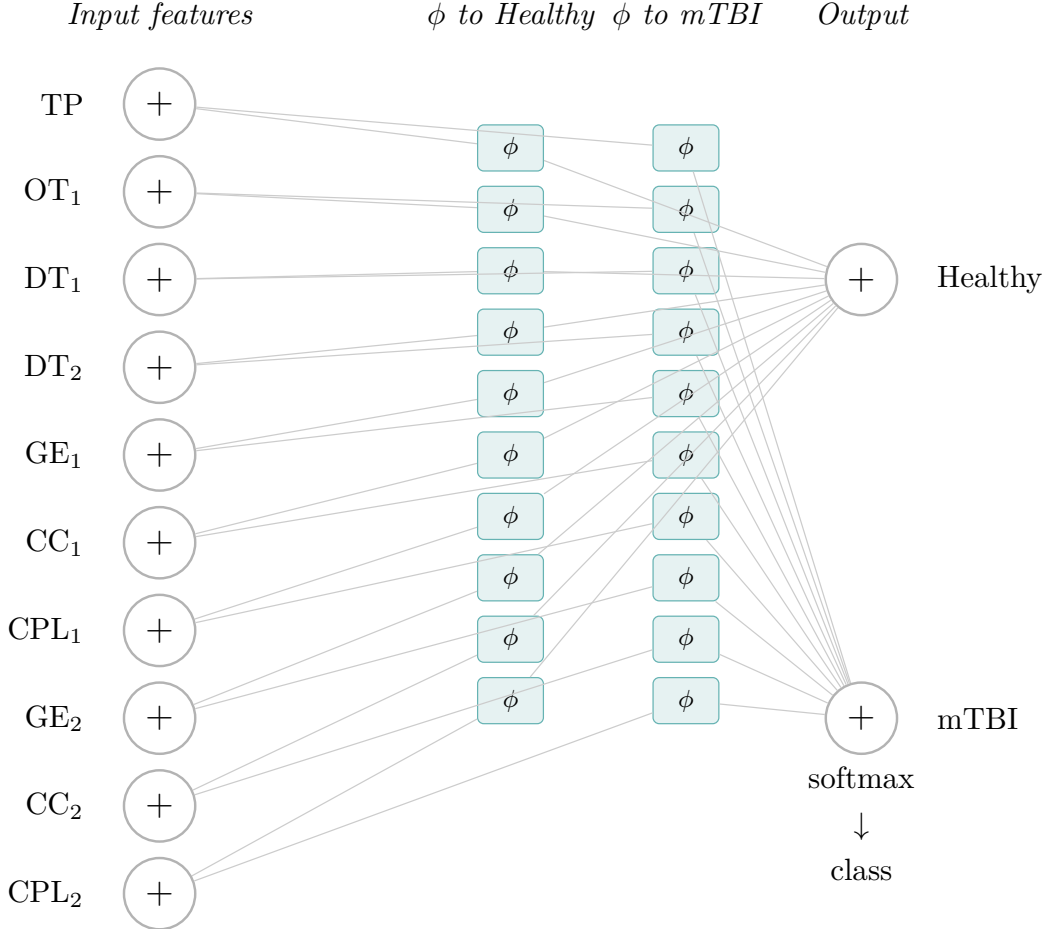


Figure 4.4: Schematic of the KAN architecture with shape $[10, 2]$. Each feature connects to both output nodes through a learnable B-spline function ϕ , with output nodes performing summation followed by softmax.

4.6 Training and Evaluation Protocol

4.6.1 Evaluation Protocol and Feature Selection

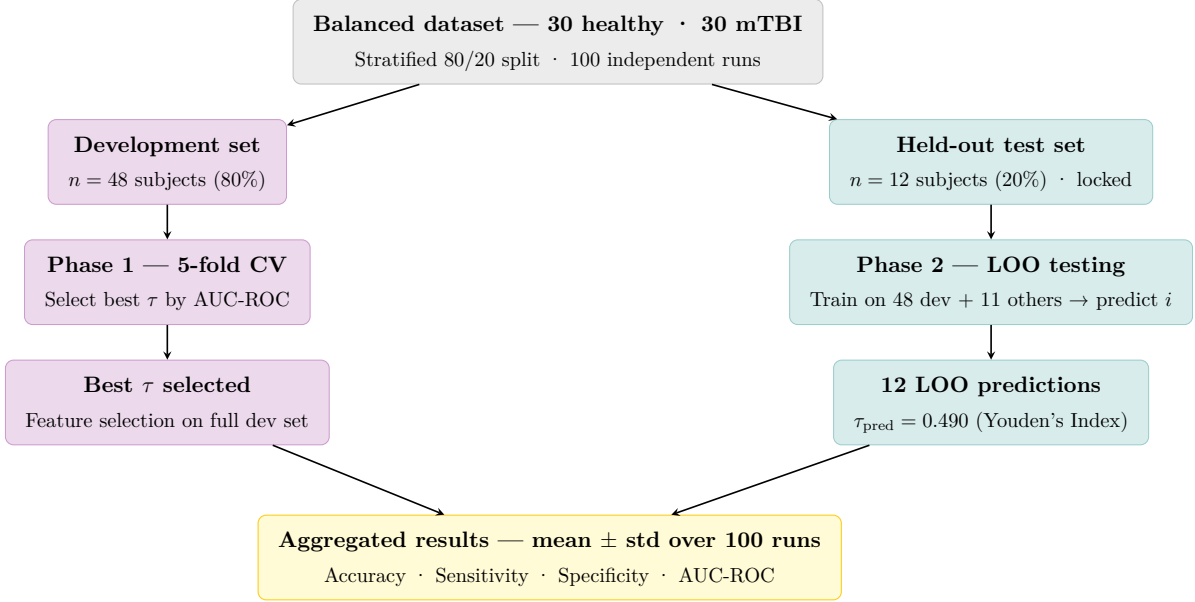


Figure 4.5: Two-phase nested evaluation protocol repeated across 100 independent runs.

Standard k -fold cross-validation is known to produce optimistically biased performance estimates on small datasets (Vabalas et al., 2019; Lewis et al., 2023). To address this, a two-phase nested evaluation protocol was adopted, repeated across $N = 100$ independent runs. At each run, the balanced dataset of 60 subjects was split into a training set (80%, $n = 48$) and a held-out testing set (20%, $n = 12$), stratified by class label.

Phase 1 — Stratified k -fold Cross-Validation. A 5-fold stratified cross-validation was performed on the training set to determine the best feature selection threshold τ based on AUC-ROC. Before each fold, a KAN of shape $[10, 2]$ was first trained to obtain feature importance scores. As described in Liu et al. (2024), each input feature’s importance is given by the mean absolute activation magnitude of the edges connected to it. Features scoring above threshold τ were kept for the final classifier. If only one feature survived, the two highest-scoring features were used instead, since KAN requires at least two inputs. The optimal threshold τ was selected from the candidate set $\{0, 0.1, 0.15, 0.2, 0.25, 0.35\}$. The pruned final classifier was then trained for 30 steps using the Adam optimizer. Following the configurations Liu et al. (2024), the regularization strength was set to $\lambda = 0.01$ and the L_1 and entropy sparsification weights to $\mu_1 = \mu_2 = 1$.

Phase 2 — Leave-One-Out Testing. The optimal threshold locked during Phase 1 was applied to define the selected feature subset. Final evaluation used a true Leave-One-Out (LOO) scheme on the 12 held-out test subjects: for each test subject i , the model was retrained on all 48 subjects from Phase 1 plus the remaining 11 test subjects (total $n = 59$ training subjects), and prediction was made on subject i alone. This maximized training data usage while keeping the prediction for each subject fully independent (Vabalas et al., 2019).

To obtain binary predictions, a decision threshold of $\tau_{\text{pred}} = 0.49$ was applied to the softmax output probabilities. This value was determined by maximizing Youden’s Index ($J = 0.453$) during Phase 1 cross-validation (Youden, 1950). Slightly lowering the classification threshold reflects the clinical reality of mTBI detection, where failing to identify an injured patient (a false negative) carries a much higher clinical cost than a false positive.

4.6.2 Overfitting Mitigation

Because of the small dataset ($n = 60$ after balancing), overfitting was a constant concern throughout the design of this pipeline, and several choices were made specifically to address it.

First, the classifier itself was kept as simple as possible: a single-layer KAN of shape $[n_{\text{feat}}, 2]$ with no hidden layers, which limited the number of parameters relative to the amount of training data. Second, the sparsification term ($\lambda = 0.01$) penalized diffuse activation functions and pruned uninformative edges (Liu et al., 2024). Third, the nested CV/LOO evaluation protocol ensured that hyperparameter selection (Phase 1) and performance estimation (Phase 2) were always performed on disjoint data splits, preventing optimistic bias (Vabalas et al., 2019; Lewis et al., 2023). Finally, the entire procedure was repeated 100 times with random data splits, yielding a distribution of performance estimates rather than a single potentially lucky result, and providing a reliable picture of the model’s stability and generalization (Lewis et al., 2023).

Empirical evidence for the absence of severe overfitting is provided by the learning curves shown in Figure 4.6. The mean cross-entropy loss over training steps was plotted, averaged across all $100 \times 12 = 1,200$ Phase 2 LOO predictions. The training loss decreased from 0.860 at step 1 to 0.719 at step 30, while the test loss decreased from 1.234 to 0.869, reaching its minimum of 0.856 at step 27. The train/test gap at the final step is 0.150, which is moderate but stable. Also, neither curve diverges, showing that 30 training steps was sufficient for the model to converge, but not long enough to cause progressive over-

fitting. Both curves remain above the random chance baseline of $\ln(2) \approx 0.693$, which reflects the difficulty of the task given the small and heterogeneous dataset rather than a failure of the model, since the classification metrics presented in Chapter ?? clearly exceed chance-level performance.

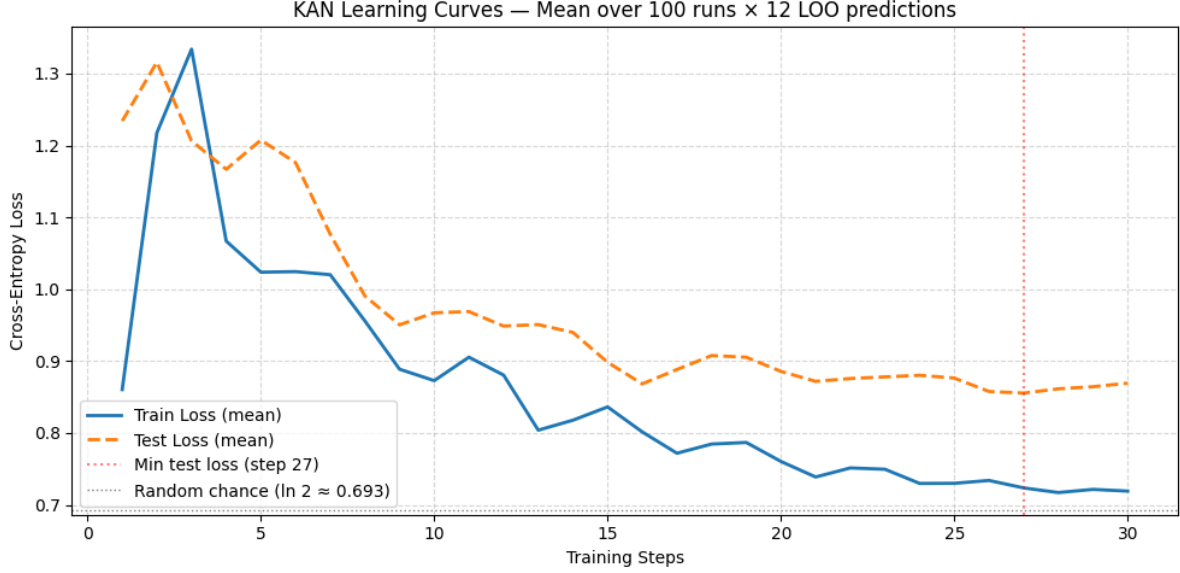


Figure 4.6: Mean cross-entropy loss over training steps, averaged across all $100 \times 12 = 1,200$ Phase 2 LOO predictions.

4.6.3 Performance Metrics

Classification performance was evaluated using metrics derived from the confusion matrix — a tabular layout that maps the model’s predictions against the actual clinical ground truth. For this binary classification task, the matrix contains four distinct cases:

- **True Positives (TP):** The model correctly classified a subject as an mTBI patient.
- **True Negatives (TN):** The model correctly classified a subject as a healthy control.
- **False Positives (FP):** The model misclassified a healthy subject as an mTBI patient.
- **False Negatives (FN):** The model misclassified an mTBI patient as healthy.

The metrics used are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Each metric provides a different perspective on the model’s behavior. Accuracy measures the overall proportion of correctly identified subjects, though it can be misleading with imbalanced datasets. Sensitivity measures the proportion of true mTBI patients correctly detected, whereas specificity measures the proportion of true healthy subjects correctly identified. Together, they capture the critical clinical trade-off between missing a real injury (a false negative) and misclassifying a healthy individual as injured (a false positive).

In addition to these threshold-dependent metrics, the Area Under the Receiver Operating Characteristic curve (AUC-ROC) was computed. The ROC curve plots sensitivity against $1 - \text{specificity}$ across all possible decision boundaries, and the area under it summarizes the model’s discriminative power, where 0.5 indicates random guessing and 1.0 indicates perfect class separation (Fawcett, 2006). To capture the model’s true classification potential independently of the chosen threshold, AUC-ROC was computed directly from the softmax probability outputs rather than the binary class labels. All metrics are reported as mean $\pm$ standard deviation across the 100 independent runs.

Chapter 5 – Results and Evaluation

5.1 Classification Performance

The diagnostic performance of the optimized network is summarized in Table 5.1, which tracks classification metrics across both the training and testing phases.

Metric	Phase 1 — 5-fold CV ($n = 48$)	Phase 2 — LOO ($n = 12$)
Accuracy	0.726 ± 0.047	0.707 ± 0.133
Sensitivity	0.739 ± 0.075	0.688 ± 0.187
Specificity	0.713 ± 0.083	0.725 ± 0.191
AUC-ROC	0.792 ± 0.045	0.760 ± 0.141

Table 5.1: Summary of Phase 1 and Phase 2 classification metrics (mean $\pm$ standard deviation across 100 independent runs).

In Phase 1, the standard deviations are relatively low, which indicates consistent performance across different splits. Also, sensitivity slightly exceeds specificity, showing the model is slightly better at identifying true mTBI patients than true healthy subjects.

In Phase 2, the wider standard deviations are expected since the test dataset only consists of 12 subjects, so a single misclassification shifts accuracy by 8.3%. The sensitivity/specificity balance also reverses slightly, suggesting the model becomes slightly more conservative in flagging mTBI when it is tested on truly unseen subjects. Nevertheless, the proximity between Phase 1 and Phase 2 AUC values proves the model generalizes well without a substantial drop in discriminative ability.

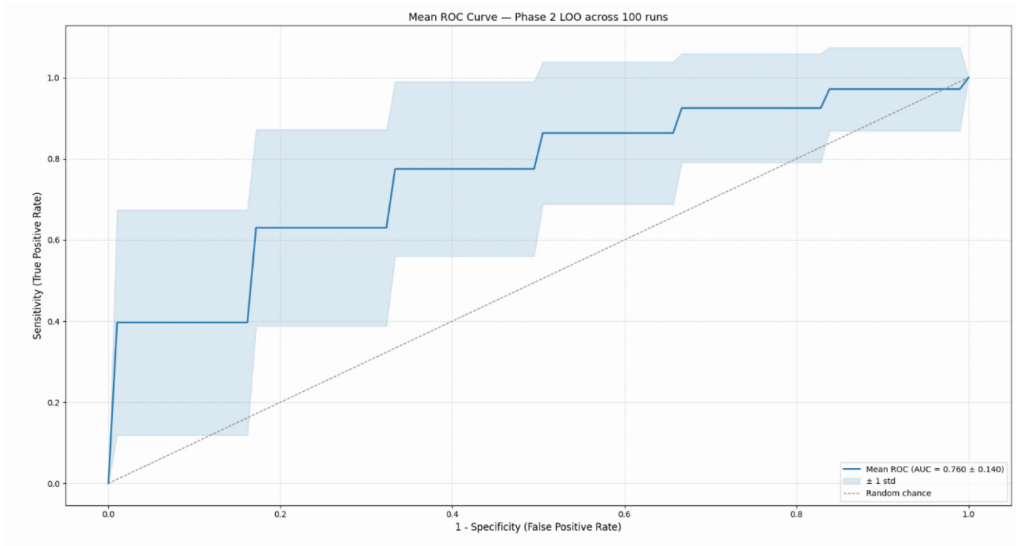


Figure 5.1: Mean Phase 2 ROC curve with a ± 1 standard deviation shaded band.

Figure 5.1 shows the mean ROC curve across all Phase 2 LOO runs, with the shaded band representing ± 1 standard deviation. The stepped shape of the curve is a direct consequence of working with a small test dataset, where sensitivity and specificity can only scale in discrete steps. The curve remaining clearly above the random chance diagonal is consistent with the discriminative performance already reported in Table 5.1.

5.2 Feature Importance

Figure 5.2 shows how each feature’s selection rate evolves during training, along with its mean importance score for all pruning thresholds. It reveals which features remain active under strict selection criteria and which drop out early.

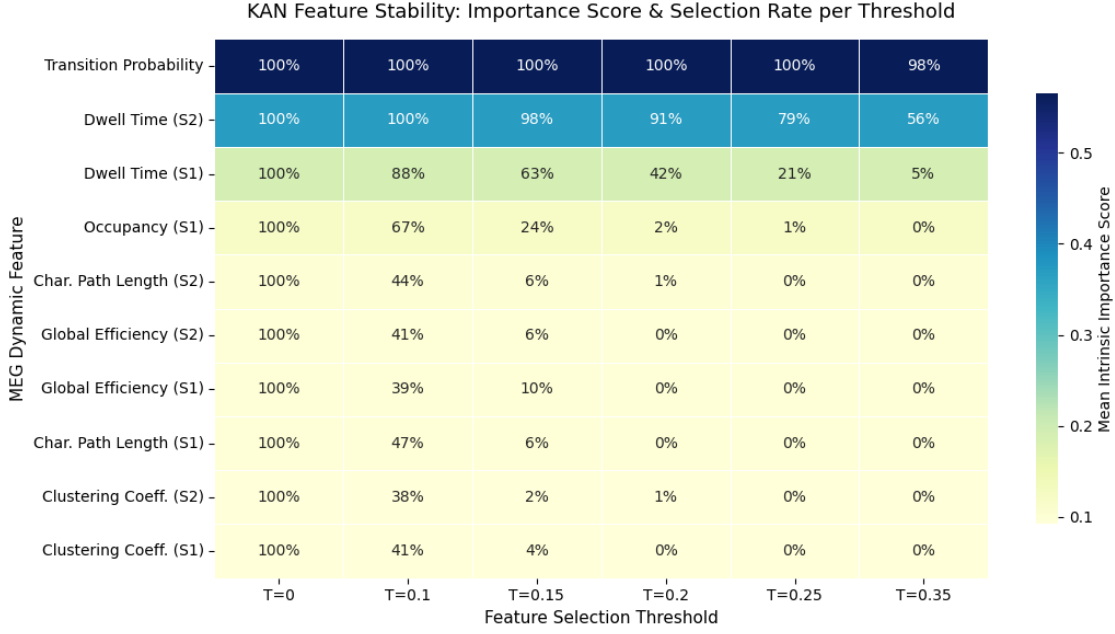


Figure 5.2: Feature stability heatmap across all 100 runs and six thresholds. Cell color represents mean importance score; cell label represents the percentage of runs in which the feature exceeded the corresponding threshold.

The contrast between temporal and graph-theoretic features is immediately visible, as the top rows (TP, DT_2 , DT_1 , and OT_1) remain colored across all thresholds. At the same time, the bottom rows (CPL_2 , GE_2 , GE_1 , CPL_1 , CC_2 , and CC_1) fade rapidly. Notably, transition probability is the only feature that maintains near-complete selection even at the strictest threshold, suggesting that it carries strong and consistent information across different data splits.

Figure 5.3 provides a cleaner overall ranking by collapsing selection rates and mean importance scores across all thresholds.

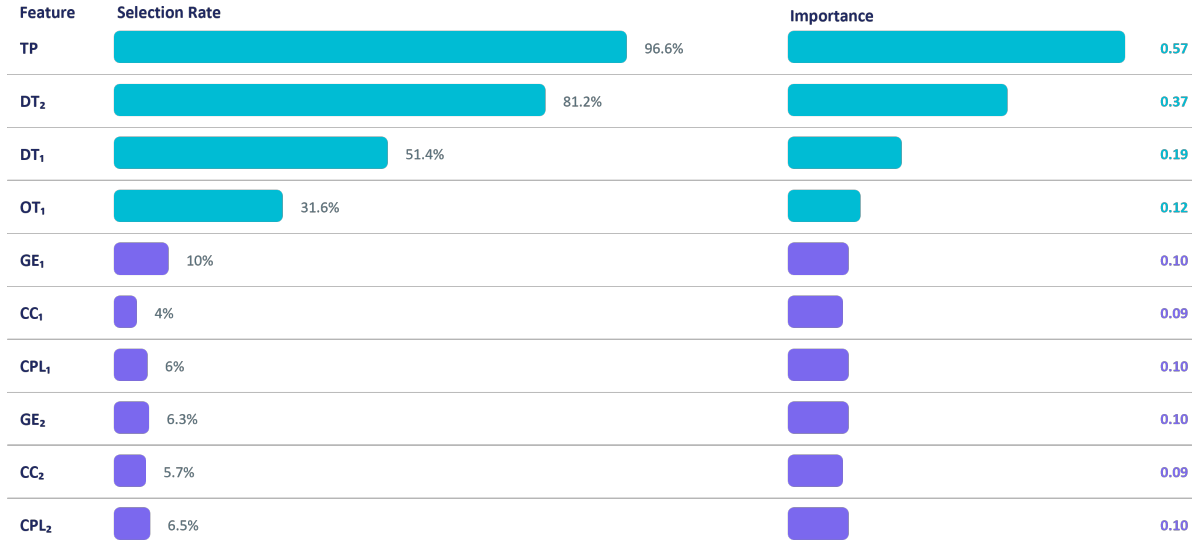


Figure 5.3: Feature selection rate and mean importance score across all 100 runs and six thresholds. Teal indicates temporal features; purple indicates graph-theoretic features.

The gap between temporal and graph-theoretic features is even more pronounced in this figure. TP, DT₂, and DT₁ achieve between two and six times higher importance scores than any graph-theoretic feature. Also, all the graph-theoretic features have similar selection rates and importance scores. This shows that none of them carries distinctly more information than the others and that their collective contribution to the classification outcome is marginal compared to the temporal dynamics.

The pattern is clear: the main discriminators of mTBI in this dataset are the temporal dynamics of microstate transitions, not the topological structure of the functional connectivity graphs. It is also worth noting that no single threshold value dominated across runs: $\tau = 0.15$ and $\tau = 0.2$ were each chosen 20% of the time, followed by $\tau = 0.35$ (19%), $\tau = 0$ (15%), and $\tau = 0.1$ and $\tau = 0.25$ (13% each). This proved that the model’s performance is not highly sensitive to the exact value of parameter τ , which is reassuring given the size of the dataset.

These results align with Antonakakis et al. (2020), who identified transition rate and dwell time as the key chronnectomic indicators of mTBI using independent statistical methods. The present findings extend this by showing that the KAN’s intrinsic feature importance mechanism successfully isolates the same biomarkers directly from the training process.

Figure 5.4 shows the full KAN architecture trained on all 10 features, where the opacity of each edge reflects its learned importance score and each small box displays the spline function associated with that feature-output pair. Consistent with the statistical findings

above, the connections from TP, DT₁, and DT₂ are visibly darker and more prominent than the rest, while several graph-theoretic features, such as CPL₂, are almost entirely transparent.

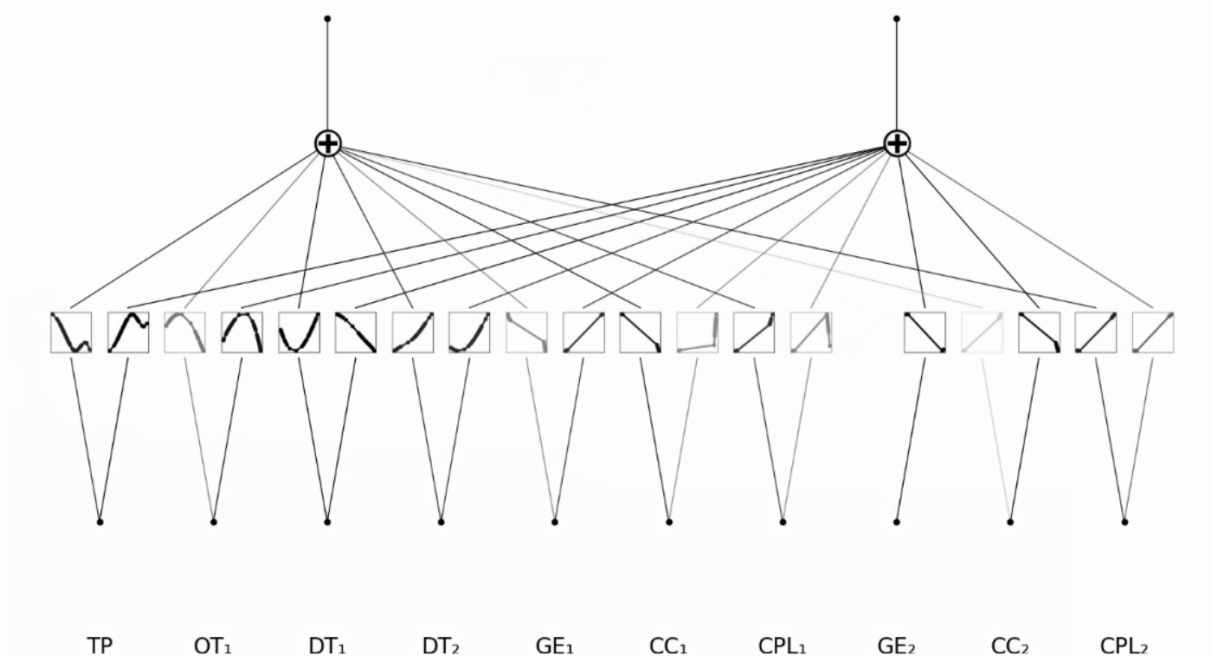


Figure 5.4: KAN architecture trained on all 60 subjects. Input nodes correspond to all 10 features, and output nodes correspond to healthy (left) and mTBI (right) classes.

The primary advantage of this representation, however, is that it further explains how each feature contributes to the classification outcome. The spline for TP shows a non-monotonic curved shape, suggesting that specific intervals of transition probability carry more information than others. DT₂ similarly appears to be sharper in specific regions, while DT₁ shows a smoother but still curved mapping. In the case of graph-theoretic features whose connections remain visible, such as GE and CC, show diagonal, almost linear splines, consistent with their lower importance scores. This means that they contribute to the decision making, but in a much simpler way than the temporal features.

5.3 Comparison with Baseline Classifiers

The same two-phase evaluation protocol that was used for KAN was applied to two standard nonlinear classifiers for comparison. These were a Support Vector Machine (SVM) utilizing a Radial Basis Function (RBF) kernel, and a standard Multi-Layer Perceptron (MLP) with a single hidden layer. For both, Phase 1 selected the best hyperparameters via 5-fold CV on the training set (SVM: $C \in \{0.1, 1, 10\}$, kernel $\in \{\text{rbf}, \text{linear}\}$; MLP:

hidden layer size $\in \{5, 10, 20\}$). Phase 2 used the same LOO validation on the 12 held-out test subjects. Table 5.2 summarizes the aggregated metrics obtained across the 100 iterations of Phase 2.

Model	Accuracy	Sensitivity	Specificity	AUC-ROC
KAN	0.707 ± 0.133	0.688 ± 0.187	0.725 ± 0.191	0.760 ± 0.141
SVM	0.746 ± 0.103	0.715 ± 0.153	0.777 ± 0.150	0.781 ± 0.121
MLP	0.758 ± 0.100	0.740 ± 0.157	0.775 ± 0.152	0.796 ± 0.123

Table 5.2: Classification performance comparison between KAN, SVM, and MLP across 100 independent Phase 2 LOO runs, reported as mean $\pm$ standard deviation.

SVM and MLP both outperformed KAN by approximately 4–5% in accuracy and 0.02–0.04 in AUC-ROC. The two baseline classifiers performed almost identically, which means that there is probably a local classification ceiling for this specific feature set at around 75–76% accuracy and 0.78–0.80 AUC-ROC. Standard deviations remained consistent across all three classifiers, confirming that the observed metric variability is mostly driven by the small test dataset rather than algorithmic instability in any of the models.

Essentially, the accuracy gap in the KAN architecture should be weighed against what it has to offer beyond a single prediction. SVM and MLP operate as traditional black boxes, outputting a classification label with no account of which features drove it. KAN, on the other hand, produces a visualizable spline function for every input-output pathway, offering a direct and interpretable mapping from brain dynamics to diagnostic outcome. The consistent emergence of transition probability and dwell time in State 2 as the most important features is not just statistically robust. It aligns with the known white matter disruption mechanism in mTBI, providing a scientifically grounded explanation that neither SVM nor MLP can offer. In a clinical setting, where understanding why a patient is diagnosed with mTBI is as vital as the diagnosis itself, KAN’s unique ability to visually justify its predictions outweighs its minor classification deficit.

Chapter 6 – Discussion

6.1 Conclusion

This thesis set out to investigate whether interpretable machine learning can extract clinically meaningful biomarkers of mTBI from resting-state MEG data without sacrificing the flexibility needed to model non-linear brain dynamics. The results provide a cautiously affirmative answer.

Transition probability and dwell time in the lower-connectivity microstate consistently emerged as the strongest biomarkers of mTBI. This aligns with Antonakakis et al. (2020) and carries a clear neurophysiological interpretation in terms of white matter disruption and destabilized network dynamics. The contribution in this work is that the KAN arrives at this conclusion intrinsically through training, without any external feature ranking, and makes the underlying decision process directly visible through its learned spline functions. The 4-5% accuracy gap observed with SVM and MLP classifiers falls within the variance of small-sample evaluation and is outweighed by this interpretive advantage, which neither baseline model can offer.

6.2 Limitations

The most significant limitation is sample size. While 60 subjects after balancing are adequate for exploratory analysis, they are not sufficient for conclusive results, and hence the performance of the model carries substantial variance throughout. The random undersampling of 30 out of the initial 49 healthy subjects varies across runs, although that variability is partly mitigated by the 100-run framework. The feature selection step introduces additional run-to-run variability in which features the model ultimately uses. This would be a practical concern in a clinical deployment where consistency of decision logic is essential. Finally, the shallow single-layer architecture, though appropriate for a small dataset, cannot capture interactions between features that may carry additional discriminative information.

6.3 Future Work

The next crucial step of this work would be validation on a larger, independent dataset, not only to strengthen the biomarker results but also to develop a deeper, more robust KAN network without the danger of overfitting. Adding multi-band chronnectomic features to the pipeline, following Antonakakis et al. (2020), could also close the performance gap while preserving interpretability. Finally, if the non-monotonic spline shapes for transition probability prove consistent across datasets, the results could lead to neurophysiologic studies to determine how diffuse axonal injury affects neural function.

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